

Lecture 4

Entity Relationship (E-R) Modeling

Outline

- Business Rules.
- M:N Relations in the presence of cardinality constraints.
- Situations.
- Multivalued attribute.

1- What are *Business Rules*, what is their source, and why are they crucial?

Business Rules

- Business rules are **precise statements**, derived from a detailed description of the **organization's operations** that define one or more of the following modeling components:
 - entities
 - relationships
 - attributes
 - connectivities
 - cardinalities
 - constraints

Business Rules (cont.)

- Because the business rules **form the basis** of the data modeling process, their precise statement is **crucial to the success** of the database design.
- Because the business rules are derived from a precise description of operations, **much of the design's success depends on the accuracy of the description of operations.**

Examples of Business Rules

- **An invoice contains one or more invoice lines, but each invoice line is associated with a single invoice.**
- **A store employs many employees, but each employee is employed by only one store.**
- **A college has many departments, but each department belongs to a single college.**

(This business rule reflects a university that has multiple colleges such as Business, Liberal Arts, Education, Engineering, etc.)

Examples of Business Rules (cont.)

- **A driver may be assigned to drive many different vehicles, and each vehicle can be driven by many drivers.**

(Note: Keep in mind that this business rule reflects the assignment of drivers during some period of time.)

- **A client may sign many contracts, but each contract is signed by only one client.**
- **A sales representative may write many contracts, but each contract is written by one sales representative.**

Business Rules (cont.)

- Keep in mind that the E-R diagrams **cannot always reflect all** of the business rules. For example, examine the following business rule:

“A customer cannot be given a credit line over \$10,000 unless that customer has maintained a satisfactory credit history (as determined by the credit manager) during the past two years.”
- This business rule describes a constraint that **cannot be shown** in the E-R diagram. The business rule reflected in this constraint **would be handled at the applications software level** through the use of a trigger or a stored procedure.

2- How are M:N relationships handled in the development of an E-R diagram in the presence of cardinality constraints?

“Assuming that the video store only rents videos and that a CUSTOMER entry will not exist unless a person coming into the video store actually rents that first tape.

CUSTOMER is *optional* to TAPE. TAPE is *mandatory* to CUSTOMER.”

- Note that this discussion includes a very brief description of the video store's operations and some business rules. The relationship between customers and tapes would thus be M:N as follows,



Customer must rent one or more tapes.

Tapes may be rented by zero or more customers.

- The M:N relationship **must be broken** up into two 1:M relationships through the use of a **bridge entity**, also known as a **Composite Entity** or **Associative Entity**.
- The composite entity, named RENTAL is composed of *at least* the primary key components of the two entities it bridges, because **the RENTAL's foreign keys must point to the primary keys of the two entities CUSTOMER and TAPE.**
- *And, because a tape that is never rented will never show up in the RENTAL entity, RENTAL has become optional to TAPE.* That's why the optional symbol has **migrated** from CUSTOMER **to the opposite side** of RENTAL.

Video Store Rental System (cont.)

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Customer must rent one or more tapes.

Tapes may be rented by zero or more customers.



- CUSTOMER **makes** RENTAL
- TAPE **is placed in** RENTAL



- Customer must make one or more rental.
- Rental must be made by one customer.
- Tape may be placed in zero or more rental.
- Rental must have one tape. X
- Rental must have one or more tapes. 12

- The **cardinalities** reflect the **rental transactions**.
- Each rental transaction, i.e., each record in the RENTAL table, will reference **one and only one customer** and **one an only one tape**.
- The (simplified!) implementation of this model may thus yield the sample data shown in the database in the following Figure.

Table name: CUSTOMER

Database name: CH3_RENTAL

	CUS_NUM	CUS_LNAME
▶	1001	Smith
	1002	Cordoza
	1003	Bianco
	1004	Jones

Table name: TAPE

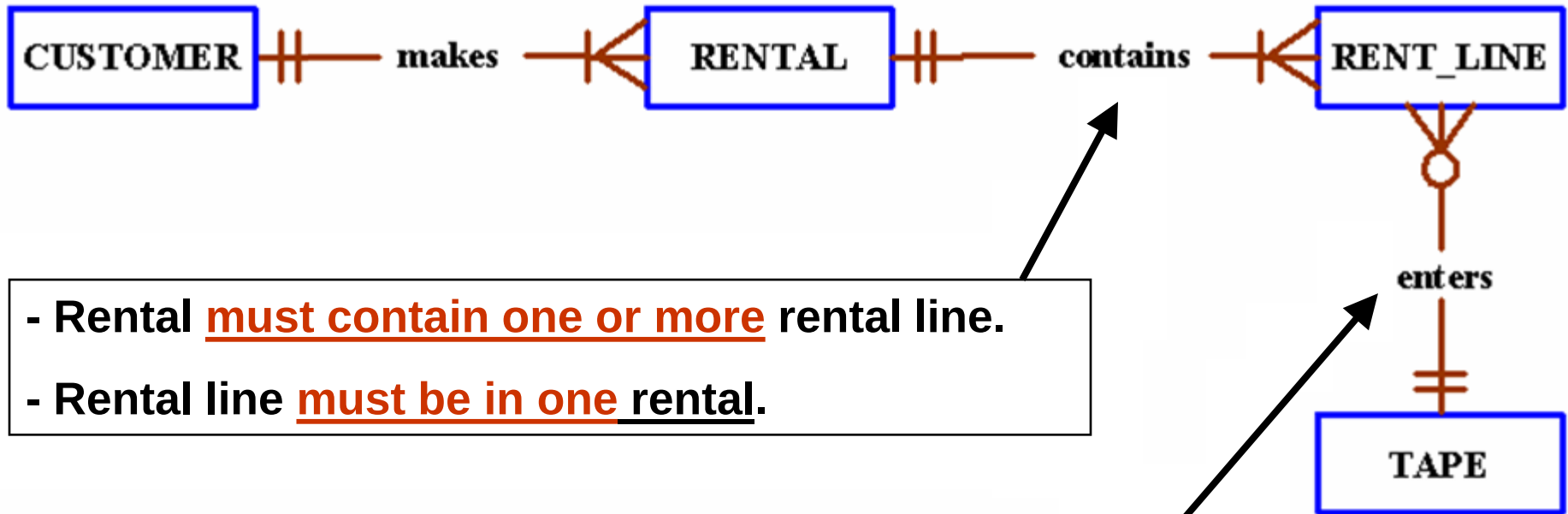
	TAPE_CODE	TAPE_COPY	TAPE_NAME	TAPE_CHARGE
▶	M3456	1	Ramblin' Tulip	\$2.50
	R2345	1	Once Upon a Midnight Breezy	\$3.99
	R2345	2	Once Upon a Midnight Breezy	\$3.99
	S4567	1	Tulips and Threelips	\$2.50
	vW1234	1	Bright Stars and Doodle Berries	\$3.00
	vW1234	2	Bright Stars and Doodle Berries	\$3.00
	vW1234	3	Bright Stars and Doodle Berries	\$3.00

Table name: RENTAL

	RENT_NUM	RENT_DATE_OUT	CUS_NUM	TAPE_CODE	TAPE_COPY	RENT_CHG_DAY	RENT_RETURN	RENT_CHARGE
▶	10050	10-Jan-2002	1002	R2345	1	\$3.99	11-Jan-2002	\$3.99
	10051	10-Jan-2002	1003	R2345	1	\$3.99	11-Jan-2002	\$3.99
	10052	10-Jan-2002	1001	vW1234	2	\$3.00	12-Jan-2002	\$6.00
	10053	10-Jan-2002	1004	M3456	1	\$2.50	11-Jan-2002	\$2.50
	10054	12-Jan-2002	1004	S4567	1	\$2.50	13-Jan-2002	\$2.50
	10055	12-Jan-2002	1001	R2345	1	\$3.99	13-Jan-2002	\$3.99
	10056	13-Jan-2002	1003	vW1234	3	\$3.00	13-Jan-2002	\$9.00
	10057	13-Jan-2002	1002	vW1234	1	\$3.00	14-Jan-2002	\$3.00
	10058	13-Jan-2002	1001	S4567	1	\$2.50	14-Jan-2002	\$2.50
	10059	14-Jan-2002	1004	M3456	1	\$2.50	15-Jan-2002	\$2.50
	10060	14-Jan-2002	1002	R2345	1	\$3.99	15-Jan-2002	\$3.99
	10061	14-Jan-2002	1003	vW1234	3	\$3.00	16-Jan-2002	\$9.00
	10062	15-Jan-2002	1001	S4567	1	\$2.50		
	10063	15-Jan-2002	1004	R2345	1	\$3.99	16-Jan-2002	\$3.99
	10064	15-Jan-2002	1004	R2345	2	\$3.99		

- Database is **not quite ready** for prime time.
- For example, its structure allows **customers to rent only one tape per rental transaction**.
- Therefore, you'd have to generate a separate rental transaction for each tape rented by a customer.
- (In other words, if a customer rents five tapes at a time, you'd have to generate five separate rentals).
- Clearly, the design would be much improved by **expanding it to include rental lines**, as is done in the following Figure.

The Updated Tape Rental Database Design



- Rental must contain one or more rental line.
- Rental line must be in one rental.

- Tape may enter zero or more rental line.
- Rental line must have one tape.

3- Situations

- A situation is a **well-defined set of circumstances** that can be described using a sufficiently complete **natural language** (Nouns, Verbs and Modifiers).
- Nouns
 - A noun is the **name** of a type of person, animal, plant, place, thing, substance or idea. (e.g. Employee)
 - A proper noun is the name of a **particular occurrence** or instance of a noun. (e.g. Mohamed, Ali, Magdy)
 - A pronoun is a word used as a substitute for a noun and that refers to a noun named or **understood** in the context in which it is used. (e.g. he, she, it)

● Nouns → Entities

Situations (cont.)

- Verbs

- A verb is a word that describes a mode of being, an association, an action or an event. (e.g Ali works in Sales)
- Verbs describe the state of nouns, and **relate the nouns** of a situation with one another.

• Verbs → Relationships

- Modifiers

- A modifier is a word that **qualifies a noun or a verb** regarding its character, quantity, degree or extent.
- Modifiers of *nouns* are called adjectives, while modifiers of *verbs* are called adverbs.

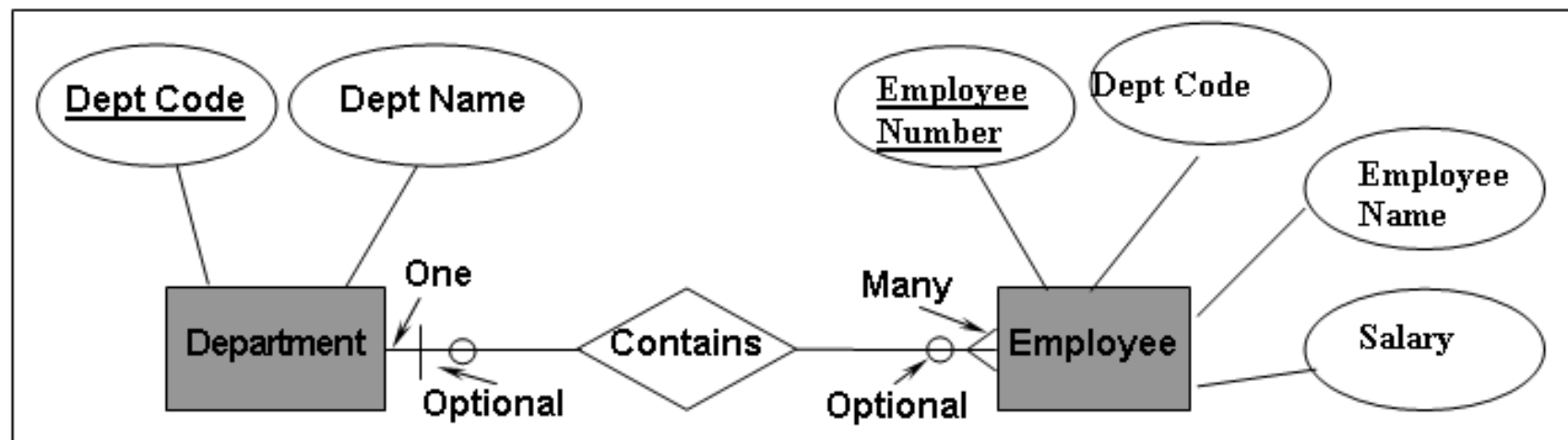
• Adjectives → Attributes of entities

• Adverbs → Attributes of associative entities

- **The customer says:**

"My company employs Drucker, Fonebone, Gaines, Newman and Jane; each is identified by a unique Employee Number. Drucker works in sales, while Fonebone and Jane works in Payroll. The others are not assigned to a department. Three other departments have been established – Research, Marketing and Graphics – but these do not have any employees at the present time. All departments are identified by two- character Department Codes. Drucker is the lowest paid employee with a salary \$12,000 a year; Newman makes twice that amount. Jane also makes \$24,000, and Gaines, the boss, makes \$12,000 more than that."

Department			Employee			
Dept Code	Dept Name		Employee Number	Dept Code	Employee Name	Salary Amount
PK, UA	NN, ND		PK, SA	FK	NN	
SA	Sales		1	SA	Drucker	12,000
PA	Payroll		2	PA	Fonebone	
RE	Research		3		Gaines	36,000
MK	Marketing		4		Newman	24,000
GA	Graphics		5	PA	Jane	24,000



Employee may be assigned to **zero or one** department.
 Department may contain **zero or more** employees.

Example 2

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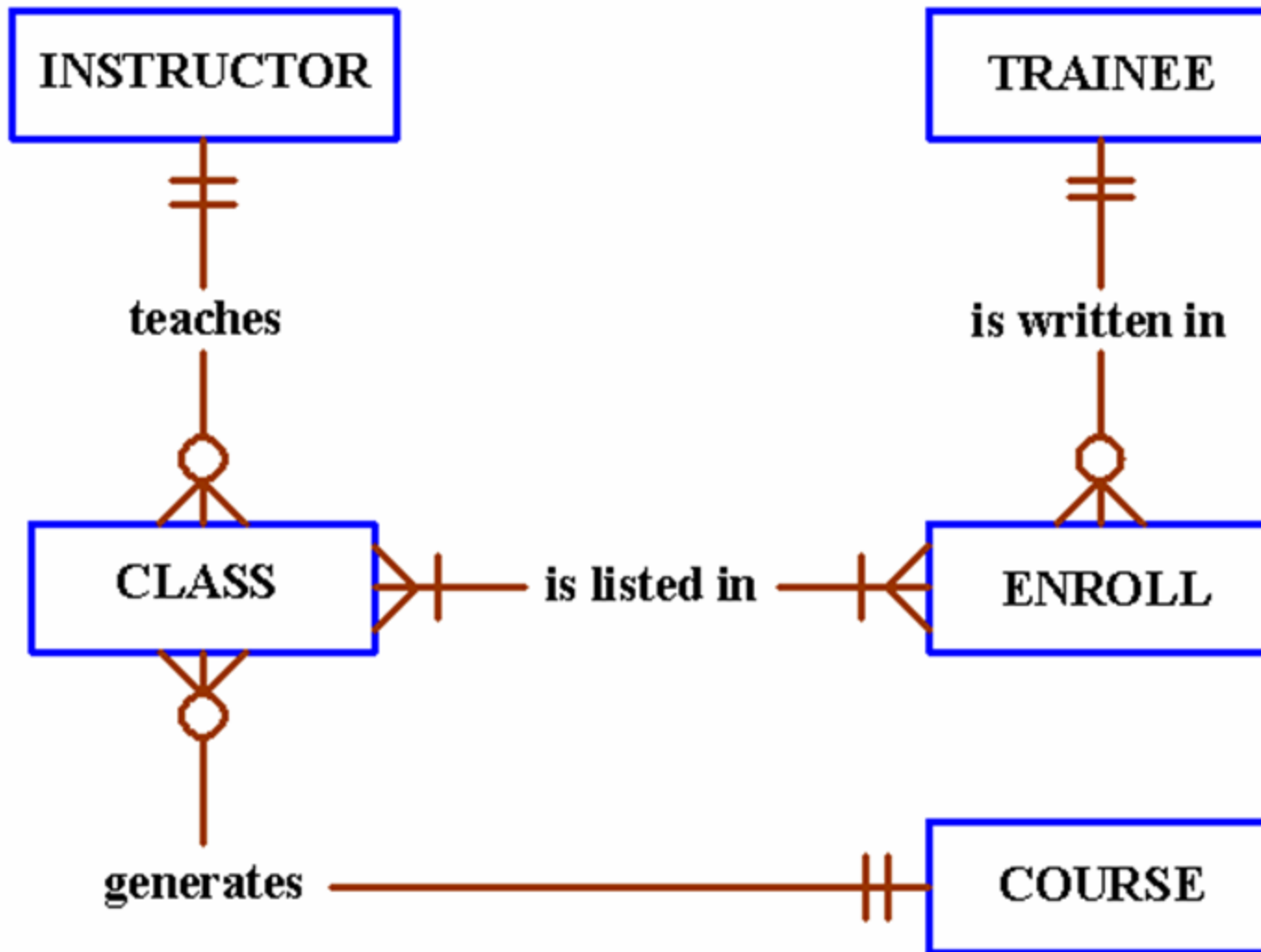
The Hudson Engineering Group (HEG) has contacted you to create a conceptual model whose application will meet the expected database requirements for its training program. The HEG administrator gives you the following description of the training group's operating environment:

“The HEG has 12 instructors and can handle up to 30 trainees per class. HEG offers five “advanced technology” courses, each of which may generate several classes. If a class has fewer than 10 trainees in it, it will be canceled. It is, therefore, possible for a course not to generate any classes during a session. Each class is taught by one instructor. Each instructor may teach up to two classes or may be assigned to do research. Each trainee may take up to two classes per session.”

Given this information, do the following:

- a. Draw the E-R diagram for HEG.
- b. Describe the relationship between instructor and course in terms of connectivity, cardinality, and existence dependence.

The Crow's Foot E-R Diagram for HEG



Write down the business rules.

4- What two courses of action are available to a designer when a multivalued attribute is encountered?

Multivalued Attributes

- The designer can **split the multivalued** attributes into its components and keep these components in the same entity.
- Examples:
 - **CAR** color is decomposed into **TOPCOLOR**, **TRIMCOLOR**, and **BODYCOLOR**.
 - **EMPLOYEE** education is decomposed into **HIGHSCHOOL**, **TWO-YEAR COLLEGE**, **FOUR-YEAR COLLEGE**, and **POST-GRADUATE**.

Multivalued Attributes (cont.)

Employee_V1

EMP_ID	Name	High_Sch	2_Coll	4_Coll	Master	PhD	Training
13658	Tarek	1/6/1985		1/6/1989			2/2/2002
32456	Mohamed	1/6/1990	1/6/1992				11/25/2003
43582	Khaled						2/25/2004
54678	Ali	1/6/1995		1/6/1999	1/25/2007		

- **Change in the number of components!!**
- **Null Values (Null is not the same as zero) !!**
- **Ability to perform queries!!**

Multivalued Attributes (cont.)

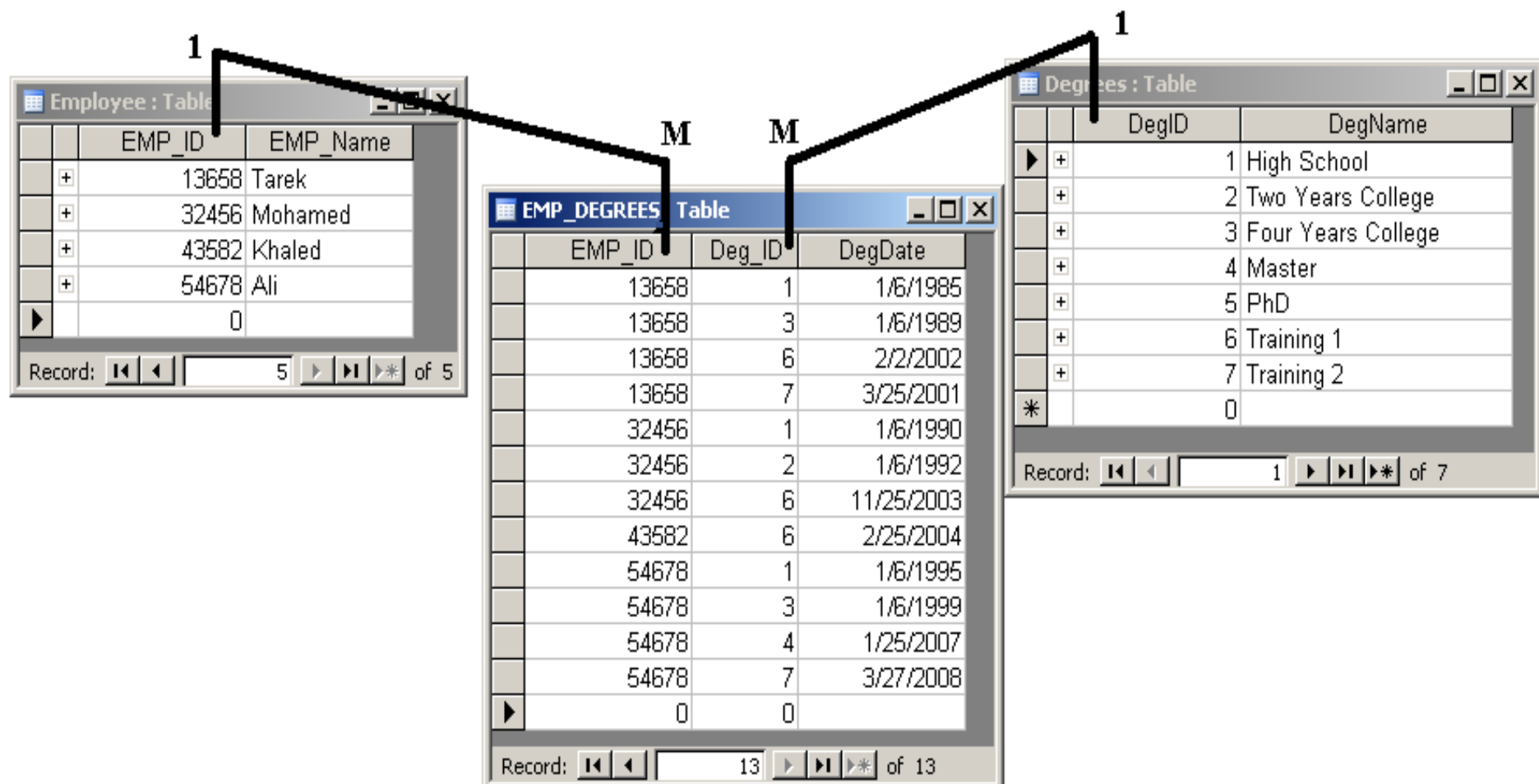
- This is the **worst possible solution**, because:
 - A **change in the number of components** forces a change in the table structure.
 - Entity occurrences within the entity (set) that **do not have all the components** included in the table structure will have **null entries**.
 - For example, if an employee only has a high school education, the TWO-YEAR COLLEGE, FOUR-YEAR COLLEGE, and POST_GRADUATE entries for that employee would be null.

Multivalued Attributes (cont.)

- A much better solution would be to **create a new entity** composed of the multivalued attribute's components and then link this new entity to the entity in which the multivalued attributes occurred originally.
- This solution is especially desirable when **the number of outcomes in the multivalued attribute is, for all practical purposes, unlimited**.
 - For example, employees classified as "technical" may have certifications in many different areas and at many different levels.

Employee_V1 : Table									
	EMP_ID	EMP_Name	High_School_Date	Two_Years_College	Four_Years_College	Master_Date	PhD_date	Training_1	Training_2 Date
	13658	Tarek	1/6/1985		1/6/1989			2/2/2002	3/25/2001
	32456	Mohamed	1/6/1990	1/6/1992				11/25/2003	
	43582	Khaled						2/25/2004	
	54678	Ali	1/6/1995		1/6/1999	1/25/2007			3/27/2008
▶	0								

Record: 5 of 5



5- Develop a data model for the following situations:

Situation 1:

"The user wants to keep track of invoices, uniquely numbered by the system, that are received from various vendors, each identified with a user-assigned vendor number. A vendor may submit more than one invoice; each may contain more than one item. A vendor may send the invoices to any branch".

Situation 2:

"A company processes orders for various kinds of parts. Each invoice is identified with a system-assigned number, as is each kind of part. An order can be for more than one kind of part, and each kind of part may appear on many orders."

Workbook Assignments

- Ex1 & Ex2 page 9
- Ex6 & Ex 7 page 32
- Ex 10 page 40